ECDLP attack graph: unestablished claims and what rests on them

THM-RP-FF-HANDBOOK proved_internal_unreviewed · impact 1 formal: kernel_checked
THEOREM-FROB-SUBSPACES-131 proved_internal_unreviewed · impact 0
RED-DEN-BOER-FIXED-CDH conditional · impact 1
RED-MAURER-AUXILIARY-CURVE conditional · impact 1
PROB-ECDLP definition
THM-RP-AFFINE-CORE proved_internal_unreviewed · impact 1 formal: lean_written_unchecked
LB-ISOGENY-GENERIC proved_internal_unreviewed · impact 0
HEUR-XT-PPVW-BOUNDED-RANK heuristic · impact 1
THM-XT-HKT-LIFT-RANK proved_external_unread · impact 1
OBS-XT-LIFT-BOUNDED-RANK conditional · impact 0
HEUR-XT-MILLER-HEIGHT heuristic · impact 1
HEUR-XT-SS-ANALYSIS heuristic · impact 0
THM-XT-MASSER-DEPENDENCE proved_external_unread · impact 1
HEUR-XT-EASY-LIFT-DEPENDENCE heuristic · impact 0
MEAS-RP-DREG-PROXY-MISS
HEUR-RP-SEMIREGULAR-DREG-PROXY refuted · impact 0
HEUR-PF-YYTK-STATISTICAL heuristic · impact 1
THM-PF-YYTK-NAIVE-LOWER-BOUND conditional · impact 0
THM-PF-REPO-KUMMER-FFD-FOUR proved_internal_unreviewed · impact 2
HEUR-PF-PKM-FFA-TRANSPOSED heuristic · impact 0
MEAS-PF-PKM-DEGREE-TABLES heuristic · impact 1
HEUR-PF-PKM-EASIER-THAN-SEMIGENERIC heuristic · impact 0
THM-PF-EP-QSP-FAMILIES
HEUR-PF-HKPYY-SPARSE-DENSITY refuted · impact 0
GAP-PF-PKM-ASYMPTOTIC-OPEN open · impact 1
GAP-PKM-SOLVING-DEGREE open · impact 0
GAP-RP-COMMON-KERNEL open · impact 1 formal: none
GAP-END-TO-END-IC-EXPONENT open · impact 0
FACT-FIELD-DLP-AFTER-PAIRING conditional · impact 1
STD-BK-SAFECURVES-TRANSFER assumed · impact 0
THM-BK-GG-SEMAEV-LEMMA2 proved_external_unread · impact 1
DEF-BK-GG-UNROLLED-SYSTEM definition
CONJ-XT-LANG-HEIGHT open · impact 1
OBSTRUCTION-XEDNI-HEIGHT proved_external
CONJ-XT-GKZ-REPRESENTATION open · impact 1
COND-XT-GKZ-ALL-EXTENSIONS conditional · impact 0
HEUR-WD-EG07-SMOOTHNESS heuristic · impact 1
COND-WD-EG-2007-L13 conditional · impact 0
HEUR-WD-SMITH-HYP1 heuristic · impact 1
HEUR-WD-SMITH-HYP2 heuristic · impact 1
HEUR-WD-SMITH-HYP3 heuristic · impact 1
HEUR-WD-SMITH-HYP4 heuristic · impact 1
COND-WD-SMITH-THM2 conditional · impact 1
HEUR-WD-DT-ASSUMPTION12 heuristic · impact 3
COND-WD-DT-RESULT13 conditional · impact 2
COND-WD-LAINE-LAUTER conditional · impact 0
COND-WD-FREY-1998 conditional · impact 5
COND-WD-DIEM-PLANE-CURVES conditional · impact 3
DEF-WD-COVER-ATTACK definition
THM-RP-S3-LINEAR-HALF-TRACE proved_internal_unreviewed · impact 1
THM-RP-SUBSPACE-PRODUCT-BOUND proved_external_unread · impact 1
COND-RP-LINEARIZED-SUBSPACE-FLOOR conditional · impact 0
HEUR-BK-GG-FFDA-DESCRIPTION heuristic · impact 1
COND-BK-GG-SEMAEV-UNROLLED-CONJ conditional · impact 0
COND-BK-GG-OVERDETERMINED-TRADEOFF conditional · impact 0
HEUR-BK-GG-BREAK-SYMMETRY heuristic · impact 0
HEUR-BK-GG-FACTOR-BASE-SIZE heuristic · impact 4
HEUR-BK-GG-DECOMP-PROPORTION heuristic · impact 3
COND-BK-GG-GAUDRY-FIXED-N conditional · impact 0
HEUR-BK-GG-RIGHT-KERNEL-RANK heuristic · impact 1
ALG-XT-GKZ-F2-367-BREAK
CLAIM-XT-SS-GENUS2-F2-367-128BIT refuted · impact 0
HEUR-XT-JOUX-PINPOINT-CORRECTED
CLAIM-XT-JOUX-PINPOINT-ORIGINAL refuted · impact 0
CEX-XT-CWZ-TRAPS
CLAIM-XT-BGJT-PROP2-ALL-P refuted · impact 0
HEUR-XT-BGJT-DESCENT heuristic · impact 2
COND-XT-GKZ-F2-1223-59BIT conditional · impact 0
CLAIM-XT-AMOR-F2-1223-128BIT refuted · impact 0
HEUR-XT-BGJT-REPRESENTATION heuristic · impact 1
COND-XT-BGJT-QUASIPOLY conditional · impact 1
FACT-GG-WALK-CLASS-WELLDEFINED
CLAIM-GG-CERTICOM-DP-NORMALIZATION refuted · impact 0
ALG-BK-MULTI-TARGET-AMORTIZED
CLAIM-BK-X962-NO-AMORTIZATION refuted · impact 0
RED-GG-FKL-AGM-CDH-DL proved_external_unread · impact 1
THM-GG-JM-AGM-TRANSFER proved_external_unread · impact 1
CLAIM-GG-FKL-AGM-TO-GGM conditional · impact 0
HEUR-BK-BRAINPOOL-CLASS-NUMBER-LIFT heuristic · impact 1
STD-BK-BRAINPOOL-REQS assumed · impact 0
CLAIM-BK-SAFECURVES-BRAINPOOL-1E6 refuted · impact 0
THM-RP-S3-ROOTS-FACTOR proved_internal_unreviewed · impact 5 formal: kernel_checked
FACT-S3-ROOTS-ARE-GROUP-LAW proved_internal_unreviewed · impact 4
BRIDGE-GENERIC-TABLE proved_internal_unreviewed · impact 3
LB-CGK-ADVICE proved_external
LB-SHOUP-DLP proved_external
ATTACK-FREY-RUECK conditional · impact 1
COND-XT-MOV-SS-SMALLCHAR conditional · impact 0
THM-XT-BK-WEIL-SUFFICES proved_external_unread · impact 6
ATTACK-MOV-PAIRING conditional · impact 5
CONCLUSION-SPECIFIC-CURVE assumed · impact 0
STANDARD-CURVE-EXCLUSIONS conditional · impact 1
STD-BK-SEC1-VALIDATION assumed · impact 0
CLAIM-BK-SAFECURVES-SEC1-20 refuted · impact 0
THM-XT-VERHEUL-DISTORTION-EXISTS proved_external_unread · impact 2
RED-PAIRING-CO-DDH conditional · impact 1
DEF-GAP-DH definition
ATTACK-CHEON-AUXILIARY conditional · impact 1
FACT-BK-KM-GENERIC-GAP-REAL proved_external
ASM-PF-PKM-UNIT-COST assumed · impact 3
HEUR-PF-PKM-RANK-DELTA heuristic · impact 3
HEUR-PF-PKM-SOLUTION-COUNT heuristic · impact 3
HEUR-PF-PKM-AUX-CURVE-SEARCH heuristic · impact 4
HEUR-PF-APS-T-EQUALS-TPRIME heuristic · impact 1
COND-PF-PKM-COST-THRESHOLD conditional · impact 3
DEF-PF-PKM-ISOGENY-BASE definition
COND-PF-APS-BEATS-PKM conditional · impact 0
COND-GB-HFE-FIRST-FALL-ESTIMATES conditional · impact 0
COND-PKM-PRIME-ALGEBRAIC conditional · impact 0
HEUR-FIRST-FALL-DEGREE heuristic · impact 5
COND-ECC2K130-FFDA-PROJECTION conditional · impact 8
CEX-NAGAO-LEMMA4-F2
CLAIM-NAGAO-LEMMA4 refuted · impact 2
COND-NAGAO-2015-COMPLEXITY conditional · impact 1
HEUR-GB-PQ-ASSUMPTION-2 heuristic · impact 2
COND-GB-PQ-SUBEXP-BINARY-ECDLP conditional · impact 0
THM-GB-HPS-FIRST-FALL-WEIL proved_external_unread · impact 3
THM-GB-PQ-FIRST-FALL-BOUND proved_external_unread · impact 1
THM-GB-HMS-NONEXISTENCE proved_external
CONJ-GB-BFS-SEMIREG-PROPORTION-F2 refuted · impact 2
HEUR-GB-SEMIREG-ASSUMPTION heuristic · impact 1
COND-BINARY-WEIL-DESCENT conditional · impact 0
CONJ-GB-FROBERG open · impact 2
THM-RP-WEIL-LIFT-SOUNDNESS proved_internal_unreviewed · impact 1
ASM-PF-KY-CURVE-CONSTRUCTION assumed · impact 2
THM-PF-KY-NPC-SMOOTH proved_external
COND-PF-KY-FFA-REDUCTIO conditional · impact 0
ASM-GG-RANDOM-PARTITION assumed · impact 5
THEOREM-RHO-THREE-PARTITION proved_external
THM-GG-MV-MIXING proved_external
THM-GG-KMT-NEAR-OPTIMAL proved_external
THM-GG-MV-NONDEGENERACY proved_external
THM-GG-KMPT-RHO-OPTIMAL proved_external
HEUR-XT-NFS-SMOOTHNESS heuristic · impact 6
ALG-XT-COPPERSMITH-F2N conditional · impact 0
COND-XT-BD-BN-100BIT conditional · impact 0
CLAIM-XT-BN3072-128BIT refuted · impact 0
ALG-XT-MSS-NORM-UPPER-BOUND conditional · impact 0
ALG-XT-NFS-L13 conditional · impact 1
ALG-XT-EXTNFS conditional · impact 2
COND-XT-BD-KEYSIZES conditional · impact 0
HEUR-WD-SUBGROUP-SMOOTHNESS heuristic · impact 3
ALG-WD-MMT-EG-COST conditional · impact 3
COND-WD-MMT-COMPOSITE-N conditional · impact 0
COND-WD-MQ-PRIME-N-FAILS conditional · impact 0
HEUR-WD-ADH-NORM-SMOOTH heuristic · impact 1
ALG-WD-ADH-1994 conditional · impact 0
ALG-STATIC-DH-AUXILIARY-POWERS conditional · impact 1
CLAIM-BK-SEC1-COFACTOR-CDH conditional · impact 0
HEUR-DECOMPOSITION-PROBABILITY heuristic · impact 8
COND-PF-APS-COST conditional · impact 0
COND-PF-HKPYY-COMPLEXITY conditional · impact 3
FACT-PF-MM-TRINOMIAL-NO-GAIN conditional · impact 0
COND-PF-EP-BETA-THRESHOLD conditional · impact 1
FACT-PF-EP-LINEARIZED-NO-GAIN conditional · impact 0
COND-RP-TWO-SUMMAND-FAMILY-LAW conditional · impact 0
HEUR-RELATION-RANK heuristic · impact 9
ALG-SPARSE-LINEAR-ALGEBRA algorithm
ALG-WD-GAUDRY-2000 conditional · impact 8
THM-WD-DIEM-GHS-ODD proved_external_unread · impact 2
THM-WD-JV-GENUS3-FAMILY proved_external
THM-WD-MOMOSE-CHAO proved_external_unread · impact 3
THM-WD-SCHOLTEN-GENUS2 proved_external_unread · impact 1
COND-WD-JV-COVER-DECOMP conditional · impact 0
ATTACK-GHS-WEIL-DESCENT proved_external
ALG-WD-THERIAULT-2003 proved_external_unread · impact 4
THM-WD-GTTD-THM1 proved_external
THM-WD-DIEM-SMALL-GENUS proved_external
COND-WD-KMPS-INEFFECTIVE conditional · impact 0
HEUR-RP-LARGE-PRIME-GRAPH heuristic · impact 1
DEF-INDEX-CALCULUS-PIPELINE definition
HEUR-RP-DESCENT-TERMINATION heuristic · impact 1
HEUR-WD-JV-WEAKNESS-INDEPENDENT heuristic · impact 1
GAP-WD-JV-ISOGENY-CONJ open · impact 0
ALG-RP-VELU-COST proved_external_unread · impact 5
RED-ISOGENY-DLP-TRANSPORT proved_external
COND-WD-GALHS-ISOGENY conditional · impact 1
COND-WD-MTW-WEAK-FIELDS conditional · impact 0
HEUR-BK-GG-ISOGENY-WALK heuristic · impact 1
THEOREM-JMV-ISOGENY-REDUCTION conditional · impact 0
THM-RP-VOLCANO-STRUCTURE proved_external_unread · impact 1
THM-GG-EQUIV-CLASS-RHO conditional · impact 2
HEUR-AUTOMORPHISM-SPEEDUP heuristic · impact 1
ALG-RHO-AUTOMORPHISM algorithm
HEUR-GG-KANGAROO-FOOTPRINTS heuristic · impact 3
ALG-GG-FOWLER-FIVE-KANGAROO conditional · impact 0
ALG-GG-GPR-FOUR-KANGAROO conditional · impact 0
ALG-KANGAROO-INTERVAL algorithm
HEUR-GG-DGM-CYCLE-RATES heuristic · impact 2
ALG-GG-BLS-BRANCHLESS-NEGATION conditional · impact 1
CLAIM-GG-BKL-SIMD-NEGATION refuted · impact 0
HEUR-RHO-RANDOM-MAPPING heuristic · impact 7
STD-BK-NIST-HALF-BITLENGTH assumed · impact 0
ALG-GG-BL-PRECOMP-RHO conditional · impact 1
STD-BK-SAFECURVES-RHO assumed · impact 1
ALG-GG-KUHN-STRUIK conditional · impact 1
HEUR-BK-GG-RHO-HEURISTIC heuristic · impact 2
STD-BK-SEC1-HARDNESS-NARRATIVE assumed · impact 0
HEUR-GG-DP-RHO heuristic · impact 1
ALG-POLLARD-RHO algorithm
THM-GG-HILDEBRAND-RADDING proved_external_unread · impact 2
HEUR-GG-TESKE-ADDING-WALK heuristic · impact 1
HEUR-SYMMETRY-PDP-SPEEDUP heuristic · impact 1
ALG-GAUDRY-FIXED-EXTENSION conditional · impact 0
HEUR-ZERO-DIMENSIONAL-PDP heuristic · impact 2
FACT-GROEBNER-COST-DEPENDENCE proved_external